

Image Transformation

Dr. Ei Moh Moh Aung
FCS



Image Translation

- Image translation in computer vision refers to the process of **transforming an image into another image** while preserving certain visual characteristics or changing them in a controlled manner.
- This transformation can involve various manipulations.

Image Translation

- **Style transfer:** Applying the **artistic style** of one image to another (e.g., making a photo look like a watercolor painting).
- **Domain adaptation:** Adapting images from one domain (e.g., satellite images) to another domain (e.g., street view images).
- **Super-resolution:** Generating a higher-resolution image from a lower-resolution one.
- **Inpainting:** Filling in missing or damaged parts of an image.
- **Image colorization:** Adding color to a grayscale image.

Image Translation: Technologies

- Deep learning:
 - Deep neural networks, particularly Generative Adversarial Networks (GANs), are widely used for image translation tasks. These networks learn the relationship between source and target images and can **generate realistic results**.
- Convolutional neural networks (CNNs):
 - CNNs excel at extracting spatial features from images, making them a suitable choice for tasks like **style transfer and domain adaptation**.
- Autoencoders:
 - These neural networks learn a compressed representation of an image and can be used for tasks like **image denoising and super-resolution**.

Image Translation: Applications

- **Artistic image editing:** Users can apply different **artistic styles** to their photos for creative expression.
- **Medical imaging:** Image translation can be used to **enhance medical images** for better diagnosis and treatment planning.
- **Autonomous vehicles:** Image translation can help self-driving cars **adapt to different weather conditions** and **environments**.
- **Video game development:** Image translation can be used to create game assets with **different styles** and **textures**.
- **Image restoration:** **Damaged or old photos** can be **repaired** and **restored** using image translation techniques.

Translation Matrix

- In OpenCV, a translation matrix is a **2x3 matrix** used to perform various kinds of image transformations, including **translating an image**, applying **affine transformations**, and **adjusting perspective**.
- It plays a crucial role in manipulating images and achieving desired visual effects.
- Here's a breakdown of its elements:
 - **2 rows**: Represent the transformation applied to the **horizontal (x) and vertical (y)** dimensions of the image.
 - **3 columns**: Represent the original coordinates of a point in the image **(x, y, 1)** and the resulting transformed coordinates after applying the matrix.

Translation Matrix

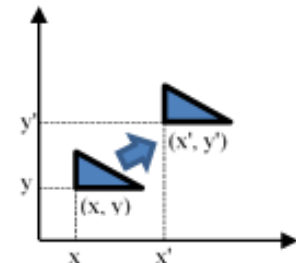
- The specific values within the matrix determine the nature of the transformation:

First row:

- $[1, 0, T_x]$: Shifts the image T_x pixels to the right while preserving its height.
- $[0, 1, T_y]$: Shifts the image T_y pixels down while preserving its width.

Second row:

- A combination of values in both columns can achieve more complex translations and tilts.



Translation Matrix

- Let's see an example of translating an image by 50 pixels to the right:
- `translation_matrix = np.float32([ [1, 0, 50], [0, 1, 0] ])`
- Here, the `[1, 0, 50]` row indicates no scaling or shearing in the horizontal direction but a shift of 50 pixels to the right. The second row `[0, 1, 0]` signifies no change in the vertical dimension.
- OpenCV provides functions like `cv2.warpAffine` and `cv2.warpPerspective` that take this translation matrix as an input and apply the corresponding transformation to the image.

Image Rotation

- Image rotation refers to the process of **turning an image around a fixed point**, resulting in a new image with the original information rearranged in a **different orientation**.
- This is a common operation in image processing used for various purposes.

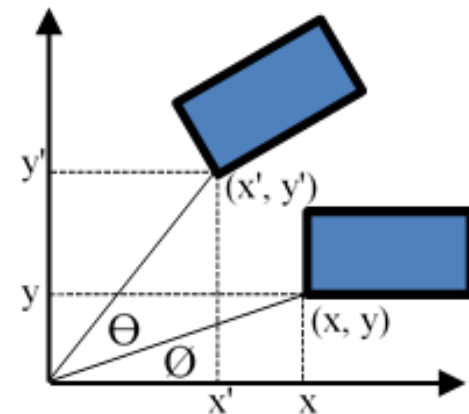


Image Rotation: Why?

- **Correcting camera tilt:** Rotating an image to compensate for the camera being tilted during capture results in a more natural-looking perspective.
- **Alignment and organization:** Images can be rotated to align features of interest or organize them in a specific layout.
- **Data augmentation:** Image rotation can be used as part of data augmentation techniques to increase the diversity of training data for machine learning models.
- **Artistic effects:** Rotating images can create interesting visual effects and distortions for artistic expression.

Image Rotation: Types

- There are two main types of image rotation:
 - **Rigid rotation**: The entire image rotates around a fixed center point, **maintaining its original shape and proportions**. This is the most common type of rotation.
 - **Affine rotation**: This involves a combination of **rotation and shear**, altering the image's shape and proportions.

Image Rotation: How

- Rotation Matrix:
 - A rotation matrix, denoted by **R**, is a **2x2 matrix** that encodes the specific rotation angle and direction.
 - The specific values within the matrix depend on the angle and axis of rotation. Here's a general form:
 - $R = \begin{vmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{vmatrix}$

where **theta** is the rotation angle in radians.

Image Rotation: How

- Applying the Transformation:
 - To rotate a point (x, y) by applying the rotation matrix R , we perform the following matrix multiplication:
$$(\text{rotated_x}, \text{rotated_y}) = R * (x, y)$$
 - This multiplication results in a new vector $(\text{rotated_x}, \text{rotated_y})$ representing the pixel's location after the rotation.

Image Rotation: How

- Rotating the Entire Image:
 - By applying the rotation matrix to each pixel's vector using the above equation, we effectively rotate the entire image around its center.
 - This process involves iterating through each pixel, performing the multiplication, and updating its coordinates based on the rotated values.

Image Resampling/Rescaling/Resizing

- Image Resizing is the process of **increasing or decreasing the size of an image**.
- What does it mean?
 - In terms of a 2-dimensional image, more specifically, it means **scaling up/down the width and height of an image** by computing the pixel values for the newer(or resized) image.

Image Resampling/Rescaling/Resizing

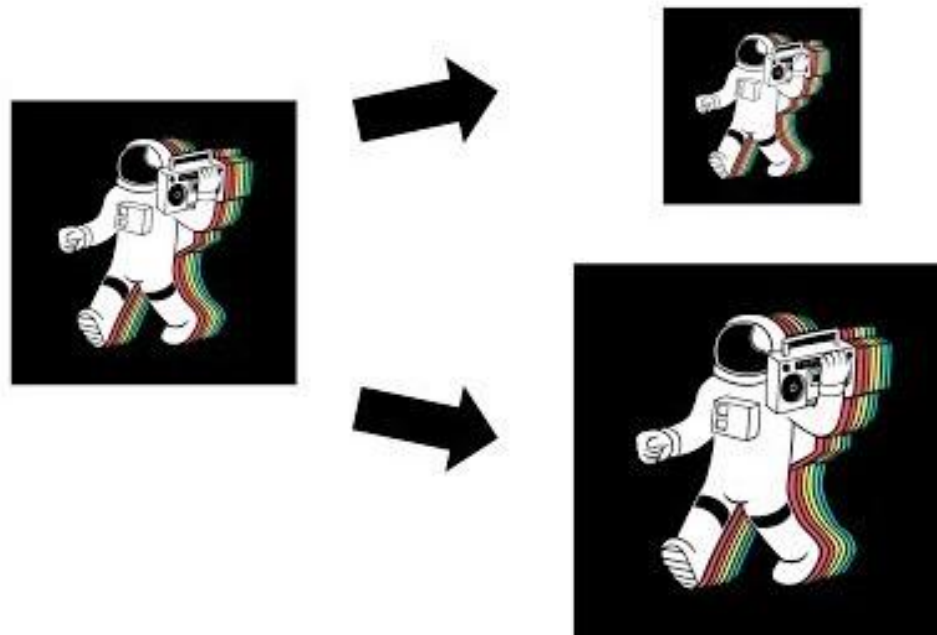


Image Resizing: How?

- **Interpolation:** This is the primary step in resizing, where missing pixels are estimated based on surrounding values. Popular methods include:
 - **Nearest neighbor:** Simplest method, assigns the value of the nearest existing pixel to the new pixel. Good for preserving sharp edges but can appear blocky.
 - **Bilinear interpolation:** Averages the values of the four nearest pixels, producing smoother results but potentially blurring image details.
 - **Bicubic interpolation:** Uses a weighted average of sixteen surrounding pixels, offering higher quality resizing but computationally more expensive.

Image Resizing: How?

- **Transformation matrices:** Linear algebra comes into play here. Resizing can be viewed as applying a transformation to the image coordinates. This can be achieved through:
 - **Scaling:** A scaling matrix **multiplies the original coordinates by a factor** (e.g., scale by 0.5 for half size), effectively shrinking or enlarging the image.
 - **Shearing:** Less commonly used, shearing matrices can tilt the image **horizontally or vertically**.

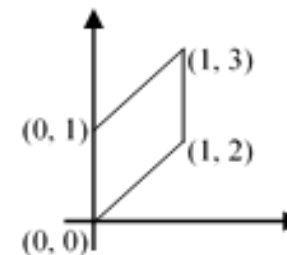
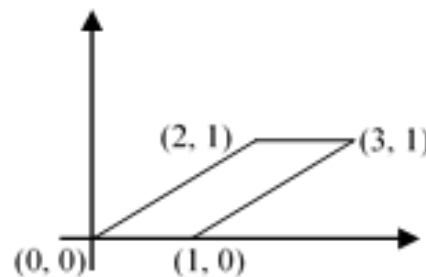
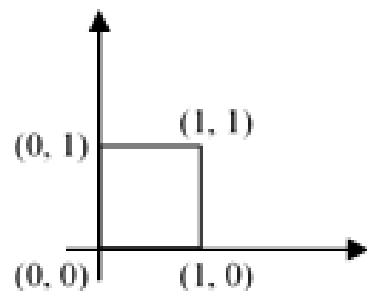


Image Flipping

- Image flipping in computer vision refers to the process of mirroring an image along a specific axis (horizontal or vertical) or both, resulting in a new image with the original information rearranged around the chosen axis.
- It's a common image manipulation technique with various applications, including:
 - Data augmentation
 - Symmetry Detection
 - Visual effects and artistic expression

Image Flipping

- Types of image flipping:
 - Horizontal flip:** Mirrors the image left to right, reversing the positions of objects and features.
 - Vertical flip:** Mirrors the image top to bottom, inverting the orientation of objects and features.
 - Diagonal flip:** Flips the image along the diagonal axis, creating a mirrored version rotated by 45 degrees.

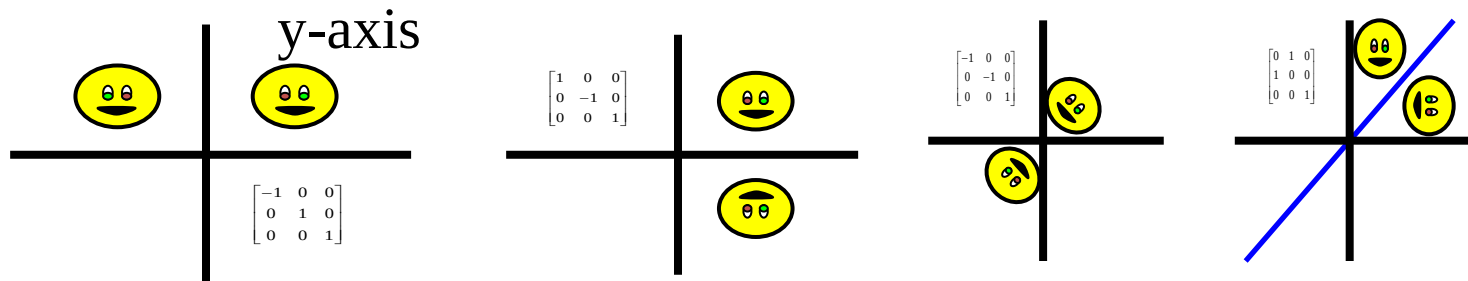


Image Cropping

- Image cropping in computer vision refers to the process of **selecting and extracting a specific region of interest (ROI)** from an image and discarding the remaining parts.



Image Cropping: Why?

- **Object detection and recognition:** After identifying an object in an image, **cropping can focus on the object** for better analysis or classification.
- **Background removal:** Cropping can isolate the subject of interest from distracting backgrounds, improving visual focus and processing.
- **Data augmentation:** Cropping images from larger ones creates multiple variations for training machine learning models, enhancing their robustness and generalizability.
- **Visual composition and editing:** Cropping can improve the composition of an image by removing unwanted elements or emphasizing specific features.

Image Cropping: How?

- **Bounding boxes:** Defining a rectangular box around the desired area is a simple and efficient method. Libraries like OpenCV provide functions for extracting ROI based on bounding box coordinates.
- **Polygons:** For irregular shapes, defining a polygon with multiple vertices offers more flexibility in selecting the ROI.
- **Masking:** Creating a mask image where the ROI is white and the rest is black allows precise extraction of the desired area.

Image Cropping: Techniques

- **Content-aware cropping:** Algorithms can analyze the image content and automatically suggest optimal cropping regions based on saliency or object detection.
- **Adaptive cropping:** The ROI can be dynamically adjusted based on specific criteria, such as maximizing the presence of objects or minimizing empty space.

Shearing

- Shearing transformation in computer vision is a specific type of geometric transformation that **distorts an image by tilting it along a specific axis**.
- This creates a slanting effect, as if layers of the image are sliding past each other in one direction.
- Types of shearing:
 - **X-shearing**: Tilts the image along the x-axis, causing vertical lines to slant.
 - **Y-shearing**: Tilts the image along the y-axis, causing horizontal lines to slant.

Shearing

- Effect on objects:
 - Shearing **alters the shape and size** of objects in the image.
 - It can be used to correct for slight rotations or introduce specific distortions for artistic effects or data augmentation.

Shearing

- Applications:
 - **Correcting text slant:** Can be used to straighten tilted text in scanned documents.
 - **Data augmentation:** Creating variations of images by introducing controlled shearing for training machine learning models.
 - **Special effects:** Used in image editing software to create artistic distortions or warping effects.

Affine Transformation

- Affine transformations modify the spatial arrangement of pixels in an image by:
 - **Scaling**: Uniformly or non-uniformly changing the size of objects.
 - **Translation**: Shifting objects to different positions.
 - **Rotation**: Rotating objects around a fixed point.
 - **Shearing**: Tilting objects along an axis, causing a slanted distortion.

Affine Transformation: Properties

- Line preservation:
 - Lines in the original image remain lines after the transformation, although their length and orientation might change.
- Parallel lines:
 - Parallel lines in the original image remain parallel after the transformation.
- Ratios preserved:
 - Ratios of distances between points are preserved, but not necessarily actual distances.

- Practical

END